ambiguous grammar - Results

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Attempt 1 of 1

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Attempt Score **8 / 11 - 72.73 %**

Overall Grade (Highest Attempt) **8 / 11 - 72.73 %**

Question 1 0 / 1 point

How many parse trees does ()() have for the grammar

S → SS | (S) | ε ?

- ○ 1
- ✗ ◉ 2
- ○ 3
- ○ 0
- ⇨ ○ infinite

Question 2 0 / 1 point

How many parse trees does ()() have for the grammar

S → (S)S | ε ?

- ⇨ ○ 1
- ✗ ◉ 2

○ 3

○ 0

○ infinite

Question 3 1 / 1 point

. Which grammar is ambiguous?

○ $S \rightarrow aSb \mid \epsilon$

○ $S \rightarrow aS \mid b$

○ $S \rightarrow abS \mid \epsilon$

✔ ◉ $S \rightarrow SS \mid a$

Question 4 0 / 1 point

Which string demonstrates ambiguity in the grammar below?
$S \rightarrow S\ S \mid a$

○ a

○ aa

➡ ○ aaa

✘ ◉ all above

Question 5 1 / 1 point

Which property is NOT sufficient to prove ambiguity?

○ Two distinct parse trees

○ Two different leftmost derivations

- Two different rightmost derivations
- ✔ Two different derivations
- None of the above

Question 6 1 / 1 point

If a grammar is ambiguous, then:

- All equivalent grammars are ambiguous
- No equivalent unambiguous grammar exists
- ✔ There may exist an equivalent unambiguous grammar
- The language itself is ambiguous

Question 7 1 / 1 point

Give the corresponding regular grammar for

S--> SS | a

- ✔ S--> aS | a
- S--> aS | epsilon
- S--> aaS | a
- S--> aaS | epsilon

Question 8 1 / 1 point

Which of the following CFGs can be transformed into a regular grammar?

- ○ S → aSb | ϵ
- ✓ ◉ S → aaS | aa
- ○ S → aSa | bSb | a| b
- ○ S → SaSb | a

Question 9 — 1 / 1 point

Given a parse tree of string **s**. **s** can be obtained from the tree by

- ○ Preorder traversal
- ○ Postorder traversal
- ○ Inorder traversal
- ✓ ◉ concatenating leaf nodes left to right

Question 10 — 1 / 1 point

Given the following parse tree:

```
   S
  / \
 a   S
    / \
   b   c
```

Which is the yield of of the parse tree?

- ✓ ◉ abc
- ○ aSSbc

- SaSbc
- aSbSc

Question 11 — 1 / 1 point

Given the following parse tree:

```
    S
   / \
  a   S
     / \
    b   c
```

Which may not be in the corresponding grammar?

- S-> aS
- S-> bc
- S ->aSbc
- more than one answer

Done